

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0004	AbsMax-Power	power-mW	min AbsMin-Power			R V	M
0x0005	PowerAdjustment-Capability	PowerAdjustCapabilityStruct	all	Q X		R V	PA
0x0006	Forecast	ForecastStruct	all	Q X		R V	PFR SFR
0x0007	OptOut-State	OptOutStateEnum	desc			R V	PA STA PAU FA CON

9.2.8.1. ESAType Attribute

This attribute SHALL indicate the type of ESA.

This attribute enables an EMS to understand some of the basic properties about how the energy may be consumed, generated, and stored by the ESA.

For example, the heat energy converted by a heat pump will naturally be lost through the building to the outdoor environment relatively quickly, compared to storing heat in a well-insulated hot water tank. Similarly, battery storage and EVs can store electrical energy for much longer durations.

This attribute can also help the EMS display information to a user and to make basic assumptions about typical best use of energy. For example, an EVSE may not always have an EV plugged in, so knowing the type of ESA that is being controlled can allow advanced energy management strategies.

9.2.8.2. ESACanGenerate Attribute

This attribute SHALL indicate whether the ESA is classed as a generator or load. This allows an EMS to understand whether the power values reported by the ESA need to have their sign inverted when dealing with forecasts and adjustments.

For example, a solar PV inverter (being a generator) may produce negative values to indicate generation (since power is flowing out of the node into the home), however a display showing the power to the consumers may need to present a positive solar production value to the consumer.

For example, a home battery storage system (BESS) which needs to charge the battery and then discharge to the home loads, would be classed as a generator. These types of devices SHALL have this field set to true. When generating its forecast or advertising its PowerAdjustmentCapability, the power values shall be negative to indicate discharging to the loads in the home, and positive to indicate when it is charging its battery.